

Data Exploration and Visualization Mini Project

Forecasting Monthly Sales Revenue

Aim

The aim of this mini-project is to explore and visualize monthly sales revenue data and to develop a forecasting model using machine learning techniques. By applying Prophet, a time series forecasting algorithm, the project seeks to predict future sales trends and help in effective business decision-making.

Algorithm

The following steps summarize the algorithm used for forecasting:

1. **Data Collection**
 - a. Import the monthly sales dataset (CSV or generated sample).
2. **Data Preprocessing**
 - a. Parse date columns and set the index as a time series.
 - b. Handle missing values or irregularities if present.
3. **Exploratory Data Analysis (EDA)**
 - a. Plot the time series to visualize trends and seasonality.
 - b. Perform statistical checks (stationarity using ADF test).
4. **Model Selection (ARIMA)**
 - a. Use Autoregressive Integrated Moving Average (ARIMA) model.
 - b. Determine ARIMA parameters (p, d, q) using ACF/PACF plots.
5. **Model Training**
 - a. Fit the ARIMA model to the training data.
6. **Forecasting**
 - a. Generate predictions for future months.
 - b. Plot actual vs. forecasted sales.
7. **Evaluation**
 - a. Compare predictions with actual values using metrics like MAE, RMSE.

Execution Steps

Prerequisites

Before running the project in **Google Colab**, ensure the following:

- Google account (for using Google Colab).
- Dataset in CSV format (or use the synthetic dataset generated in the code).
- Required Python libraries installed:
 - numpy
 - pandas
 - matplotlib
 - seaborn
 - statsmodels

Steps to run in Google Colab:

- Open Google Colab.
- Create a **new notebook**.
- Upload your dataset (if available) or use the sample dataset provided in the code.
- Copy-paste the **complete code** (already structured in cells).
- Run each cell sequentially:
 - **Cell 1:** Import libraries & load dataset.
 - **Cell 2:** Data preprocessing & summary.
 - **Cell 3:** Data visualization.
 - **Cell 4:** Stationarity check & ACF/PACF plots.
 - **Cell 5:** ARIMA model training.
 - **Cell 6:** Forecasting & visualization of results.
 - **Cell 7:** Evaluation metrics.
- Observe the plots and results after each step.
- Modify dataset or parameters as needed to experiment further.

Applications

- **Retail Industry** – Forecasting product sales for inventory management.
- **E-commerce** – Predicting monthly revenue for marketing strategies.
- **Finance & Investment** – Revenue trend analysis for decision-making.
- **Manufacturing** – Demand prediction for supply chain optimization.
- **Hospitality & Tourism** – Seasonal forecasting of bookings and revenue.

CODE:-

```
!pip install prophet

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from prophet import Prophet

from sklearn.metrics import mean_absolute_error, mean_squared_error

sns.set(style="whitegrid")

# -----

# Step 1: Create/Load Dataset

# -----

# Create sample monthly sales dataset (you can replace with your own CSV file)

date_rng = pd.date_range(start='2018-01-01', end='2023-12-01', freq='MS')

np.random.seed(42)

sales = np.random.randint(20000, 50000, size=len(date_rng)) + np.linspace(0, 10000, len(date_rng))
```

```
df = pd.DataFrame({'Month': date_rng, 'Sales': sales})

df.set_index('Month', inplace=True)

print("Sample Sales Data:")

print(df.head())


# -----

# Step 2: Visualize Data

# -----

plt.figure(figsize=(12,6))

plt.plot(df.index, df['Sales'], marker='o', linestyle='-')

plt.title("Monthly Sales Revenue Over Time")

plt.xlabel("Month")

plt.ylabel("Revenue")

plt.show()


# -----

# Step 3: Prepare Data for Prophet

# -----

# Prophet requires columns 'ds' (date) and 'y' (value)

prophet_df = df.reset_index().rename(columns={"Month": "ds", "Sales": "y"})

train_df = prophet_df.iloc[:-12]

test_df = prophet_df.iloc[-12:]

print("Train Data Range:", train_df['ds'].min(), "to", train_df['ds'].max())

print("Test Data Range:", test_df['ds'].min(), "to", test_df['ds'].max())
```

```
# -----  
  
# Step 4: Build & Fit Prophet Model  
  
# -----  
  
model = Prophet()  
model.fit(train_df)  
  
# Forecast into future including test period  
  
future = model.make_future_dataframe(periods=12, freq='MS')  
  
forecast = model.predict(future)  
  
print("\nForecast sample:")  
  
print(forecast[['ds', 'yhat', 'yhat_lower', 'yhat_upper']].tail())  
  
  
# -----  
  
# Step 5: Plot Forecast vs Actual  
  
# -----  
  
plt.figure(figsize=(12,6))  
  
# Training data  
  
plt.plot(train_df['ds'], train_df['y'], label='Training Data')  
  
# Actual test data  
  
plt.plot(test_df['ds'], test_df['y'], label='Actual Sales (Test)', color='black')  
  
# Forecast predictions  
  
plt.plot(forecast['ds'], forecast['yhat'], label='Forecasted Sales', color='red')  
  
# Confidence intervals  
  
plt.fill_between(forecast['ds'], forecast['yhat_lower'], forecast['yhat_upper'],  
                color='pink', alpha=0.3)  
  
plt.legend()  
  
plt.title("Forecast vs Actual Sales")
```

```
plt.show()

# -----
# Step 6: Model Evaluation
# -----

pred_test = forecast.set_index('ds').loc[test_df['ds'], 'yhat']

mae = mean_absolute_error(test_df['y'], pred_test)

rmse = np.sqrt(mean_squared_error(test_df['y'], pred_test))

print(f"\nMean Absolute Error (MAE): {mae:.2f}")

print(f"Root Mean Squared Error (RMSE): {rmse:.2f}")

# -----
# Step 7: Forecast Future Sales (Next 12 Months)
# -----

# Forecast 12 months beyond dataset

future2 = model.make_future_dataframe(periods=24, freq='MS') # 12 test + 12 future

forecast2 = model.predict(future2)

# Extract only future period (last 12 months)

future_forecast = forecast2.set_index('ds').iloc[-12:][['yhat', 'yhat_lower', 'yhat_upper']]

plt.figure(figsize=(12,6))

plt.plot(df.index, df['Sales'], label='Historical Sales')

plt.plot(future_forecast.index, future_forecast['yhat'], label='Future Forecast', color='green')

plt.fill_between(future_forecast.index, future_forecast['yhat_lower'], future_forecast['yhat_upper'],

                color='lightgreen', alpha=0.3)

plt.legend()

plt.title("Future 12-Month Sales Forecast")
```

```
plt.show()

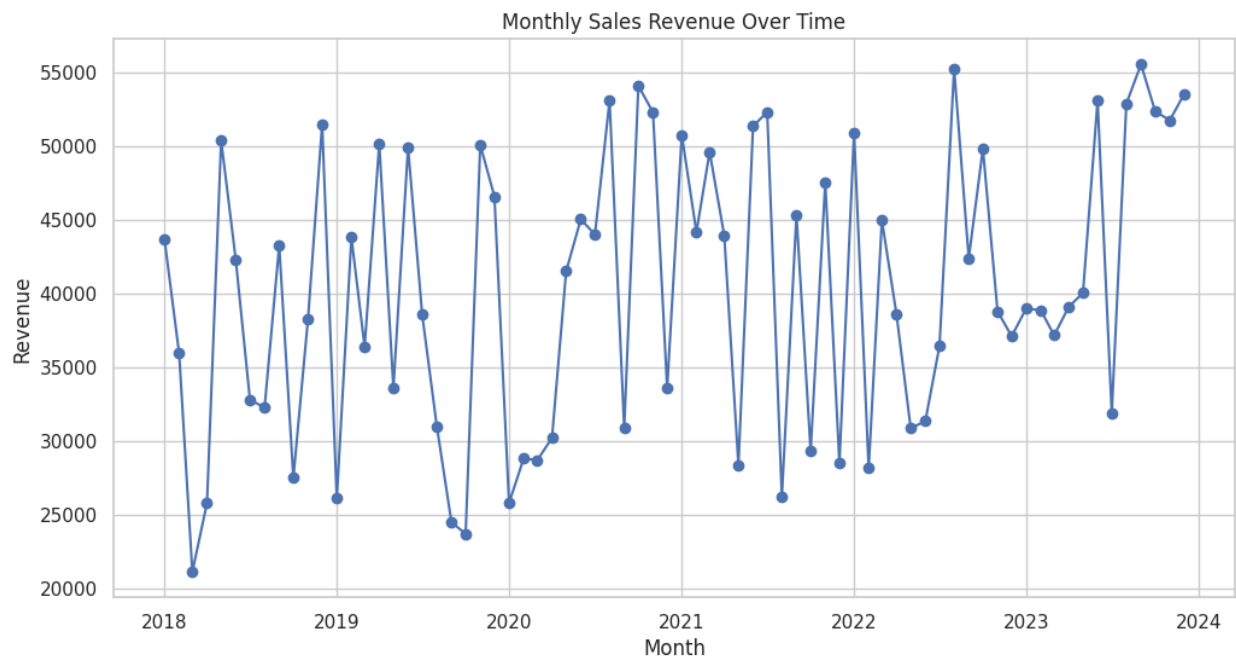
print("\nFuture Forecast (next 12 months):")

print(future_forecast)
```

OUTPUT:

Sample Sales Data:

Month	Sales
2018-01-01	43654.000000
2018-02-01	35935.845070
2018-03-01	21141.690141
2018-04-01	25812.535211
2018-05-01	50365.380282



Train Data Range: 2018-01-01 00:00:00 to 2022-12-01 00:00:00

Test Data Range: 2023-01-01 00:00:00 to 2023-12-01 00:00:00

INFO:prophet:Disabling weekly seasonality. Run prophet with weekly_seasonality=True to override this.

INFO:prophet:Disabling daily seasonality. Run prophet with daily_seasonality=True to override this.

DEBUG:cmdstanpy:input tempfile: /tmp/tmpcmpixj2z/slpno9ll.json

DEBUG:cmdstanpy:input tempfile: /tmp/tmpcmpixj2z/yccw4yie.json

DEBUG:cmdstanpy:idx 0

DEBUG:cmdstanpy:running CmdStan, num_threads: None

DEBUG:cmdstanpy:CmdStan args: ['/usr/local/lib/python3.11/dist-packages/prophet/stan_model/prophet_model.bin', 'random', 'seed=37821', 'data', 'file=/tmp/tmpcmpixj2z/slpno9ll.json', 'init=/tmp/tmpcmpixj2z/yccw4yie.json', 'output', 'file=/tmp/tmpcmpixj2z/prophet_modelpc8cei74/prophet_model-20250816144002.csv', 'method=optimize', 'algorithm=newton', 'iter=10000']

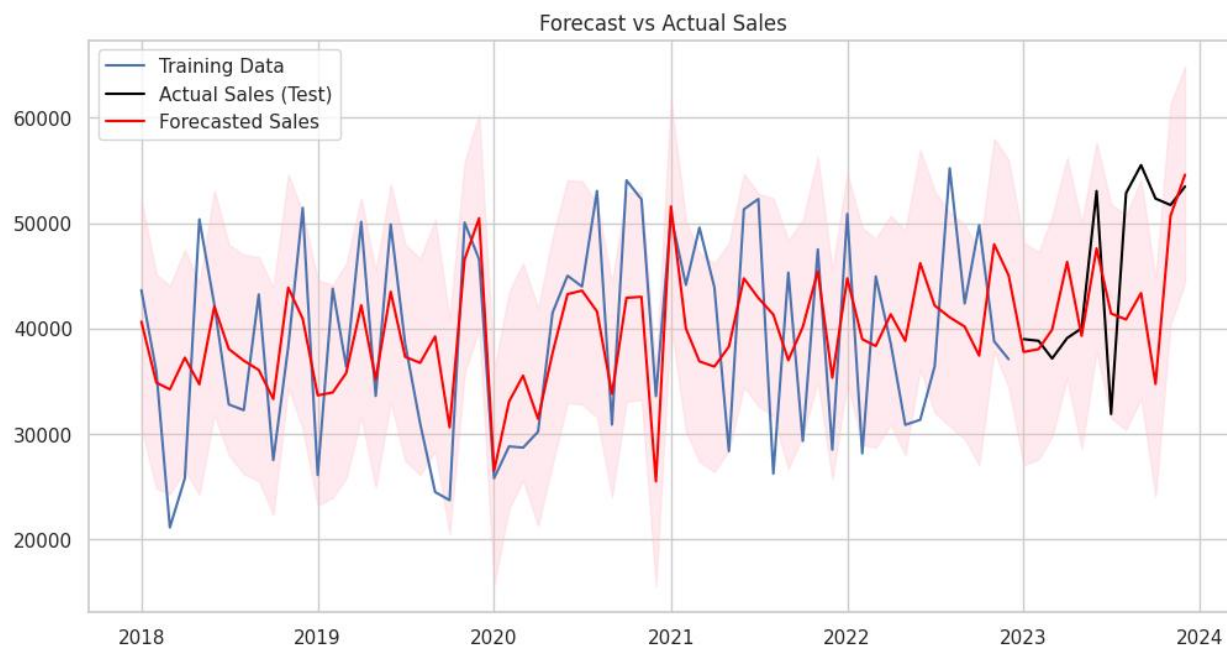
14:40:02 - cmdstanpy - INFO - Chain [1] start processing

INFO:cmdstanpy:Chain [1] start processing

14:40:02 - cmdstanpy - INFO - Chain [1] done processing

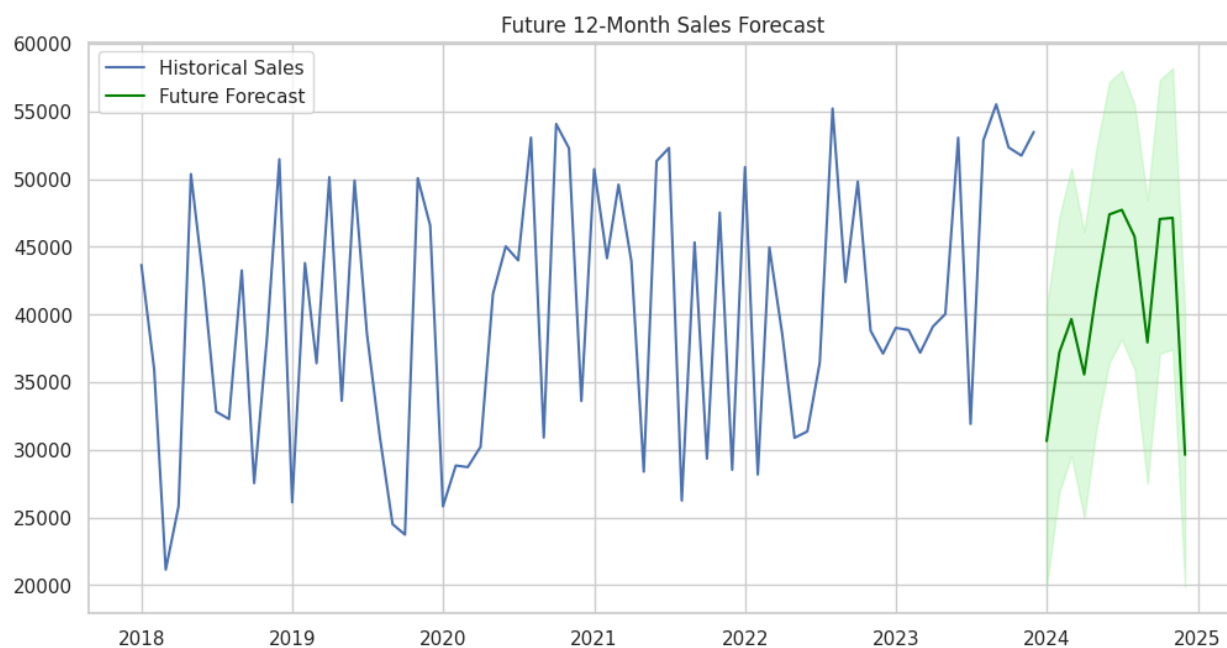
INFO:cmdstanpy:Chain [1] done processing

	ds	yhat	yhat_lower	yhat_upper	
67	2023-08-01	40871.432484	30408.999051	50777.721229	
68	2023-09-01	43374.929826	33399.260295	54117.636449	
69	2023-10-01	34756.502100	24054.304851	44700.098911	
70	2023-11-01	50692.050744	40393.614285	61300.141642	
71	2023-12-01	54598.086020	44396.228784	64884.722678	



Mean Absolute Error (MAE): 5960.26

Root Mean Squared Error (RMSE): 8089.70



	yhat	yhat_lower	yhat_upper	
ds				
2024-01-01	30640.383720	19965.954662	40480.901450	
2024-02-01	37200.673057	26902.473825	47212.294119	
2024-03-01	39651.969328	29562.884102	50799.674482	
2024-04-01	35561.906807	24940.191346	46075.734260	
2024-05-01	41808.468604	31681.374097	52347.478341	
2024-06-01	47383.830020	36455.661355	57179.661355	
2024-07-01	47723.343455	38172.408659	58004.535200	
2024-08-01	45741.632486	35926.684439	55453.769139	
2024-09-01	37930.140407	27492.835182	48441.334398	
2024-10-01	47041.687631	37108.379245	57307.658737	
2024-11-01	47134.562396	37401.044877	58187.380952	
2024-12-01	29627.967553	19864.865522	39897.531843	

Conclusion

The project successfully demonstrates the process of data exploration, visualization, and forecasting using Prophet for monthly sales revenue. The model not only predicts future values but also provides confidence intervals to account for uncertainty.

Through evaluation metrics such as MAE and RMSE, the accuracy of predictions was validated. The future 12-month forecast highlights upcoming sales trends, making this approach highly beneficial for businesses to make data-driven decisions, optimize resources, and improve financial planning.